

GROVE, Australia

WMO: 959770

Lat: 42.986S

Lon: 147.074E

Elev: 213

StdP: 14.58

Time Zone: 10.00 (AUS)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.2	30.9	29.5	23.7	52.0	31.4	25.9	46.5	22.1	50.3	18.8	49.4	1.8	30	0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.1	83.6	63.3	79.2	62.5	75.3	61.4	67.4	75.9	65.0	74.6	63.1	71.8	10.8	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.2	94.0	67.9	62.0	83.8	66.9	59.9	77.6	65.6	31.9	75.5	30.1	75.2	28.7	72.1	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.0	17.1	15.1	26.6	96.2	1.7	3.4	25.4	98.6	24.4	100.6	23.5	102.5	22.3	104.9		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.4	60.6	60.0	57.7	52.8	49.1	45.1	44.6	46.3	49.2	51.9	54.8	57.6		
	DBStd	7.61	6.08	5.64	5.80	5.38	5.23	4.69	4.44	4.93	4.98	5.44	5.47	5.41		
	HDD50	727	1	1	6	29	82	162	178	139	73	40	12	4		
	HDD65	4670	167	160	238	368	494	598	631	580	475	409	311	239		
	CDD50	1608	328	280	244	113	53	13	12	25	48	97	156	239		
	CDD65	78	29	19	12	2	0	0	0	0	1	2	4	9		
	CDH74	1147	410	243	171	18	0	0	0	0	3	32	101	169		
	CDH80	337	138	77	45	2	0	0	0	0	0	5	21	49		
(14) Wind	WSAvg	5.9	7.3	6.0	5.7	4.9	5.1	4.1	4.8	5.6	6.5	6.9	6.7	7.2		
(15) Precipitation	PrecAvg	29.3	2.0	1.9	2.0	2.4	2.3	2.2	3.0	3.1	2.8	2.7	2.6	2.4		
	PrecMax	43.3	5.7	6.4	5.8	13.3	6.2	4.9	8.8	6.8	6.9	7.2	5.6	7.1		
	PrecMin	19.6	0.4	0.2	0.3	0.2	0.7	0.4	1.1	0.9	1.0	1.3	0.2	0.2		
	PrecStd	5.3	1.3	1.6	1.3	2.1	1.4	1.0	1.7	1.6	1.4	1.4	1.2	1.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.3	89.4	85.0	76.2	68.0	61.9	60.5	65.5	70.6	78.5	83.5	85.5		
		MCWB	66.5	65.6	63.8	62.5	54.8	53.8	50.0	52.5	54.9	58.2	60.7	62.6		
	2%	DB	83.4	80.7	78.5	71.4	64.3	59.3	57.8	61.2	65.4	72.1	76.6	78.7		
		MCWB	64.2	63.2	62.6	58.1	54.4	51.3	49.1	50.8	53.4	55.7	62.1	61.9		
	5%	DB	78.1	75.6	73.5	66.9	61.7	57.3	55.8	58.4	62.3	67.4	70.7	73.2		
		MCWB	62.6	62.6	62.7	57.5	52.9	49.9	48.1	49.4	52.2	55.1	58.8	60.1		
	10%	DB	73.5	71.9	69.2	63.6	59.2	55.0	53.9	55.9	59.2	63.4	66.5	69.2		
		MCWB	60.4	60.5	59.6	55.3	51.7	48.8	47.2	48.5	50.4	53.1	56.9	58.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.6	68.4	73.3	66.2	58.7	57.0	52.7	54.4	61.5	61.8	66.9	66.6		
		MCDB	78.2	78.6	74.0	68.5	62.5	59.5	57.3	59.9	62.6	69.1	70.7	76.8		
	2%	WB	66.8	65.9	66.3	60.9	55.6	52.7	50.4	52.4	55.7	58.8	63.3	64.0		
		MCDB	78.9	77.1	73.6	68.9	61.6	57.2	56.0	59.1	63.0	68.7	72.0	73.0		
	5%	WB	63.9	63.7	63.2	58.0	53.9	50.9	49.0	50.7	53.1	56.0	60.6	61.9		
		MCDB	74.1	74.1	71.2	64.5	60.3	55.8	54.7	57.4	59.8	64.9	69.1	70.0		
	10%	WB	61.7	61.6	60.7	55.9	52.3	49.4	47.8	49.0	51.1	53.7	57.8	59.7		
		MCDB	71.0	69.9	68.5	62.5	58.3	54.6	53.3	55.2	58.2	62.0	65.7	68.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	22.1	21.4	20.1	19.3	16.6	16.5	15.8	16.5	17.7	19.3	19.4	20.0		
		MCDBR	31.9	29.1	27.7	25.6	19.0	18.8	17.3	19.6	22.4	27.4	28.8	28.2		
		MCWBR	14.7	14.0	15.3	14.8	11.8	11.0	10.0	10.6	12.9	14.4	15.2	14.2		
	5% WB	MCDBR	26.7	25.5	24.5	23.0	18.0	16.2	15.0	17.2	19.8	24.0	25.5	24.2		
		MCWBR	15.2	13.6	16.3	15.2	12.3	11.0	9.9	10.9	13.7	14.1	15.8	14.8		
(40) Clear-Sky Solar Irradiance	taub		0.346	0.333	0.333	0.315	0.307	0.294	0.296	0.309	0.332	0.336	0.335	0.340		
	taud		2.509	2.562	2.555	2.585	2.588	2.604	2.603	2.569	2.490	2.494	2.517	2.515		
	Ebn at Noon		311	306	291	274	253	246	255	272	286	302	312	314		
	Edh at Noon		35	31	29	24	20	18	19	23	30	33	34	35		
(44) All-Sky Solar Radiation	RadAvg		2028	1735	1280	865	527	419	482	720	1086	1531	1840	2029		
	RadStd		150	108	73	50	36	35	25	50	55	85	142	120		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)		+0.58	N/A	N/A	+1.22	N/A	N/A	N/A	-180	+191	+25

Nomenclature: See separate page